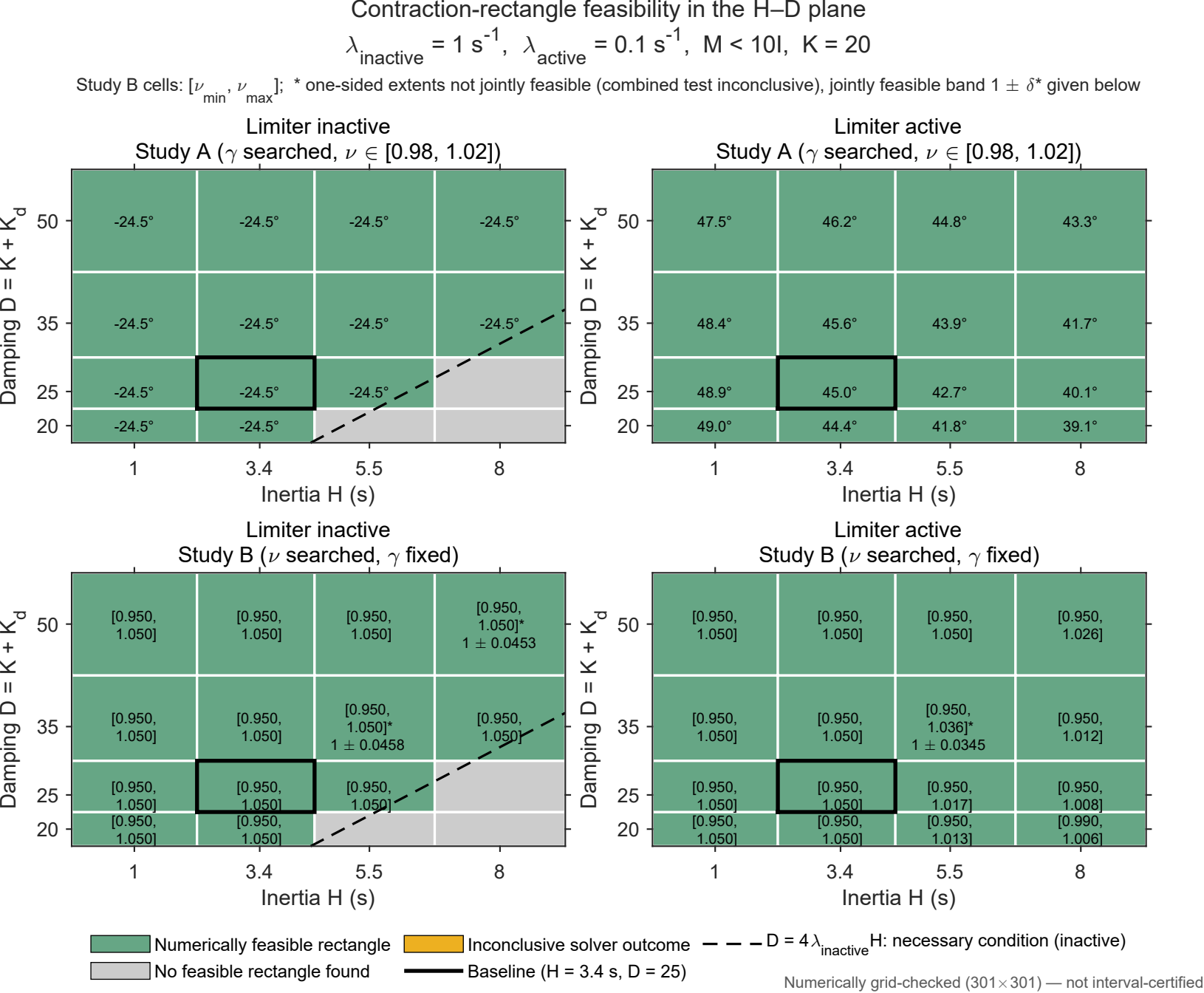
Contraction-rectangle feasibility in the H–D plane
$\lambda_{inactive}$ = 1 s^{-1}, λ_{active} = 0.1 s^{-1}, M < 10I, K = 20
Study B cells: [ν_{min}, ν_{max}]; * one-sided extents not jointly feasible (combined test inconclusive), jointly feasible band 1 ± δ* given below
Limiter inactive
Study A (γ searched, $\nu \in$ [0.98, 1.02])
Limiter active
Study A (γ searched, $\nu \in$ [0.98, 1.02])
Limiter inactive
Study B (ν searched, γ fixed)
Limiter active
Study B (ν searched, γ fixed)
Damping D = K + K_d
Inertia H (s)
1
3.4
5.5
8
20
25
35
50
-24.5°
47.5°
46.2°
44.8°
43.3°
48.4°
45.6°
43.9°
41.7°
48.9°
45.0°
42.7°
40.1°
49.0°
44.4°
41.8°
39.1°
[0.950, 1.050]
[0.950, 1.050]* 1 ± 0.0453
[0.950, 1.050]* 1 ± 0.0458
[0.950, 1.036]* 1 ± 0.0345
[0.950, 1.026]
[0.950, 1.012]
[0.950, 1.017]
[0.950, 1.008]
[0.950, 1.013]
[0.990, 1.006]
Numerically feasible rectangle
Inconclusive solver outcome
D = 4 $\lambda_{inactive}$H: necessary condition (inactive)
No feasible rectangle found
Baseline (H = 3.4 s, D = 25)
Numerically grid-checked (301×301) — not interval-certified